Fourteen integer unknowns and smaller separating norms for the consistency of ZFC

AI-assisted research prepared for Vladimir Reshetnikov's ProveIt repository

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Abstract

We compare the seven recent Diophantine presentations of the blank computation of K215 and combine their strongest independent reductions. The principal new construction uses fourteen unrestricted integer unknowns. A Pell orbit removes one order constraint, and a cofinal choice of the Jones power-of-two base removes two more. We also obtain smaller twelve-positive-variable norms, a lower-degree nine-positive-variable equation, a fifty-integer quartic, and a cheaper moderate-degree twenty-four-integer equation. The smallest positive-variable form still has nine unknowns. Every equivalence below is an ordinary mathematical argument formalizable in ZFC, using the identified compiler and machine-calibration results. Finite replay checks supplement these arguments; they are not new end-to-end Lean or Rocq proofs. No consistency decision, global minimum, or universal representation record is claimed.

1 Scope, baseline, and what "simpler" means

The review is pinned to ProveIt commit `8c6eec9d19ef80538c8e61ff6e0180040583446a`. The source collection is Con(ZFC) in Diophantine form, specifically packages 24–30. The previous delivery is package 27. There is no single order on simplicity: domain, number of unknowns, equation count, total degree, arithmetic-circuit cost, coefficient height, and proof length must be distinguished. In particular, nine positive unknowns cannot simply be read as nine unrestricted integers.

All operation counts charge binary addition, subtraction, and multiplication, including multiplication by numerals and squaring. Reuse of a previously computed value and comparison of two system outputs are free. Computing the fixed coefficients is not charged to evaluation of a polynomial in its unknowns. Thus these counts are not bit-complexity estimates. Huge constants are supplied either as actual binary integers or exact bounded arithmetic recipes, never as unspecified existential parameters.

Source	Main contribution	Use in this synthesis
24, integer24	Degree 1069 positive norm; sharp signed weights	Halved carrier and leading cancellation; signed weight bound
25, hybrid12	Degree 1081 at 116 operations; smaller tuple	Comparison; whole-tuple discipline retained
26, ratio12	Ratio-bounded roots and cheaper systems	Comparison of domains and free-comparison costs
27, previous delivery	Fast nine-variable circuits; degree 1081/120; integer lifts	Shared-power and coefficient-alias evaluation; signed and low-degree proofs
28, compact norms	Structured majorants and signed finalizer	Lower-degree nine-variable equation
29, simpler forms	Explicit 17-integer Jones specialization	Square stack, intersective merger, separated three-root norm
30, quartic31	31-variable quartic; trace/remainder norm; 111-operation product	Smaller Jones parameter; register elimination; scalar norm evaluator

The collection's corrected reconciliation is included as `sources/README.md`. It is an intake comparison, not an independent proof of every delivered assertion. Our comparison distinguishes identical polynomials from equal-looking circuits in different coefficient tuples. In particular packages 25 and 27 share two coefficient integers but not the entire tuple. Likewise replacing a tuple in package 18's circuit changes its fixed polynomial.

Selected new frontiers

Put $L = 5^{32}$. The selected equations below all satisfy, in their stated domains, ZFC $\vdash$ (no solution) $\leftrightarrow$ Con(ZFC).

Domain and unknowns	Degree	Operations	Equations
$14\mathbb{Z}$	$813056L + 2752$	1524	1
$16\mathbb{Z}$	$406528L + 10008$	1537	1
$9\mathbb{Z}_{>0}$	$92324L + 296$	1500	1
$9\mathbb{N}$	$92324L + 296$	1505	1
$12\mathbb{Z}_{>0}$	1061	120	1
$12\mathbb{Z}_{>0}$	1065	119	1
$12\mathbb{Z}_{>0}$	1069	117	1
$12\mathbb{Z}_{>0}$	1081	109	1
$12\mathbb{Z}_{>0}$	1052	114	3
$12\mathbb{Z}_{>0}$	1072	103	3
$24\mathbb{Z}$	1936	161	1
$24\mathbb{Z}$	1920	156	3
$50\mathbb{Z}$	4	197	1
$50\mathbb{Z}$	2	111	29

The two selected twelve-variable systems use an explicit transport-coordinate shift; their variable domain is unchanged. The 109-operation product is literally the source30 polynomial evaluated more cheaply. The other new separating-weight formulas are different polynomials with proved equivalent zero existence. The fourteen-variable degree is 18 930 435 180 664 062 500 000 002 752, so it is not a low-degree competitor. The nine-positive degree is 2 149 585 634 469 985 961 914 062 796.

2 One coherent coefficient and theorem interface

To avoid any new compiler-rebinding obligation, the main new families use the entire original compact943 tuple retained by package 30:

$$B_* = 2^{5^{10}}, \quad \beta = B_* - 1, \quad K_* = D_C + B_* D_R, \quad M_C, \quad M_F = M_F^{\text{native}} + B_* - 1.$$

The fourth coefficient is already shifted. The four raw, unsigned little-endian integers occupy 5,312,269 bytes and are included with their recipe and hashes. The smaller 2^{5^9} tuples of packages 25 and 27 remain valid competing coefficient choices; retaining the larger established tuple here is deliberate, not a new height record.

The shared coefficient reconstructor reproduces all four source hashes and checks 69,230 support/convolution identities. These finite statements authenticate the recipe and bytes, not its unbounded compiler theorem. That theorem, its canonical completeness properties, and the finitary calibration of the literal K215 blank run remain mathematical dependencies. The resulting proof chain is

$$\begin{aligned} &\text{finite ZFC refutation} \iff \text{K215 blank halt} \\ \iff &\text{source Diophantine solution} \iff \text{new solution.} \end{aligned}$$

Negation gives the requested ZFC-provable equivalence to Con(ZFC). It does not prove either side of that equivalence.

For the Jones families we use package 30's normalized quartic in 31 witnesses. It removes the register G from the old 32-variable system using

$$G = q(q - F) - C, \qquad L_0 G = (q - F)(N - q) - C L_0, \qquad N = q(L_0 + 1).$$

Every substitution is polynomial, and the converse restores $G > 0$ on positive zeros using the transport equation. The resulting 22 quadratic residuals give a normalized quartic by summing their squares after shifting all 31 positive coordinates by one. The dummy input is fixed to one, so the Jones parameter is $\nu = 31$, not 9.

The 402 nonzero factorial-weighted coefficient records and their index recipes are included. Writing $L = 5^{32}$ and $s_* = 45{,}933{,}757$, the explicit index has

$$z_* = 2^{s_*}, \qquad y_* = \sum_{|\alpha| \le 4} (z_* + c_\alpha)(2z_*)^{L - \sum_{j=0}^{31} \alpha_j 5^j}.$$

There are $\binom{36}{4} = 58{,}905$ positions, including zero coefficients. The bound certificate gives $|c_\alpha| + 1 < z_*$, and hence $y_* < (2z_*)^{L+1}$. The integer z_* is materialized; y_* and the larger base-scale constant used below are exact recipes, not materialized payloads. New finalizers are ordinary derivations from the Jones ratio interface, not literal instances of Lean's theorem about its particular `nineValue` polynomial.

3 Fourteen unrestricted integer unknowns

We combine package 30's quartic in 31 positive unknowns, package 27's fast Jones evaluation, and package 29's signed square-stack and intersective finalizer. Two new restricted-completeness arguments and a Pell orbit remove three of package 29's guards. The resulting fixed polynomial has

14 integer unknowns, $\quad \deg F_{14} = 813056 \cdot 5^{32} + 2752, \quad$ 1524 binary operations.

A lower-degree alternative has 16 integer unknowns, degree $406528 \cdot 5^{32} + 10008$, and 1537 operations. These are ordinary, unformalized reductions, not optimality claims or a proof of consistency. The complete derivation and exact source boundaries are in `signed/SIGNED_PROOF.md`.

3.1 Fixed index and the exact polynomial

Use the *whole* compact943 coefficient tuple of package 30 and its normalized quartic $P_{31}(x,\mathbf{n}) = \sum_j r_j(\mathbf{n}+1)^2$. The dummy x is fixed to 1. Put $L = 5^{32}$ and

$$c_\alpha = (4-|\alpha|)!\prod_j \alpha_j!\,[\mathbf{X}^\alpha]P_{31}, \qquad z = 2^{45933757},$$

$$e(T) = \sum_{|\alpha|\le 4}(z+c_\alpha)T^{L-\sum_{j=0}^{31}\alpha_j 5^j}, \qquad y = e(2z).$$

The 402 nonzero coefficient records are in `signed/upstream/COMPACT_CONSTANTS.json`. Unlisted coefficients are zero; the index certificate proves $|c_\alpha|+1 < z$. Consequently $y < (2z)^{L+1}$. Define the additional fixed integer

$$\kappa = 45933758(L+1)+1, \qquad a = 2^\kappa.$$

Thus κ is odd and $a > y$. This enormous coefficient is given by a bounded recipe, *not materialized*. The same applies to y and the weighted coefficients. Fixed leaves are not existential variables.

The complete list of unknowns is

$$(\varepsilon, g, h, s, w, \phi, i, j, z_1, z_2, z_3, z_4, v, u) \in \mathbb{Z}^{14}.$$

Set $b = a\varepsilon^2$, $\omega = w^2$ and define the following *polynomial abbreviations*, none of which is an extra unknown:

$$B = b^5, \quad q = B^L, \quad \ell = \sum_{r=1}^{31} B^{5^r}, \quad \lambda = \sum_{r=0}^{4L-1} B^r, \quad \theta = B - 2z, \quad N = q^{16}, \quad G = 32bB^{5^{31}}.$$

$$\begin{aligned} R = {} & \left[g + e(B)q^3 + \ell q^5 + \left(2(e(B)-z\lambda)(1+B+g)^4 + \lambda B + \lambda B q^4\right) q^7\right](N^2-N) \\ & + \left[q^3 - b\ell + \ell + \theta\lambda q^3 + (B-2)q^8\right](N^2-1). \end{aligned}$$

$$U = N^2\omega, \quad Y = N^2 s, \quad M = RY, \quad P = 2M^2U, \quad K = R+1+h(P-1), \quad A = M(U+1),$$
$$B_1 = 2R+1, \quad W_1 = b\omega, \quad C = B_1 + W_1 + \phi, \quad J = 4A-5, \quad Q = 3W_1C - 2(W_1^2-1),$$
$$D = (A^2-1)C^2+1, \quad E = iJDC^2, \quad F = (A^2-1)E^2+1,$$
$$G_1 = A + F(CD+1-A), \quad H = B_1 + jC, \quad I = (G_1^2-1)H^2+1,$$
$$a_0 = (P^2-1)K^2+1, \quad a_1 = DFI, \quad m = K^2G - 4(C-KY)^2G - 8gK^2,$$
$$D_0 = JF, \qquad T_0 = QF + J(H-C).$$

Put $(d_1,d_2,d_3,d_4) = (3g-1, 3s-1, 3\phi+2, 3m-1)$ and

$$C_0 = 1, \qquad B'_j = 1 + 4d_jC_{j-1}^2z_j^2, \qquad C_j = C_{j-1}B'_j \ (1\le j\le 4), \qquad C_* = C_4.$$

Finally set

$$Z = \varepsilon w i z_1z_2z_3z_4, \quad S = ZD_0^5, \quad r = T_0 + D_0v,$$
$$T = (r^3 - D_0^2r - D_0^3)(r^2+23D_0^2), \quad w_1 = C_*+1, \quad w_2 = w_1(P^2+K^2+2), \quad X = Su - T,$$
$$A_0 = S^2C_*, \quad A_1 = S^2w_1^2a_0, \quad A_2 = S^2w_2^2a_1,$$
$$V = X^2 + A_2 - A_0 - A_1, \qquad W = V^2 + 4X^2A_2 - 4A_0A_1.$$

The requested equation is the single integer polynomial equation

$$\boxed{F_{14} = W^2 - 16V^2X^2A_2 = 0.}$$

Its division-free expanded arithmetic DAG is `signed/generated/signed14.json`. The exported names `residue` and `normroot` mean v, u.

3.2 The signed Jones interface and a removable guard

The imported Jones ratio theorem, on its original domain $b = y+1+\varepsilon_0$, $\varepsilon_0, \phi \geq 0$ and $g, h, s, \omega, i, j \geq 1$, identifies quartic solvability with

$$a_0, a_1 \in \mathrm{Sq}(\mathbb{Z}), \qquad D_0 \mid T_0, \qquad m > 0. \tag{JR}$$

For every integer assignment $D_0 \neq 0$: $J = 4A - 5 \neq 0$, and $F = 0$ would force $A = 0, E = \pm 1$, although E would be divisible by 5. Replacing a nonzero i by $|i|$ leaves (JR) unchanged since only E^2 occurs in its consumers.

Package 29's proof of automatic h positivity does not use j. For completeness, assume $b > y$, $g, s, \omega \geq 1$, $\phi \geq 0$, a_0 square and $m > 0$. The margin gives

$$K \neq 0, \quad g < 4bB^{5^{31}}, \quad 4(C - KY)^2 < K^2.$$

Writing $t = 5^{31}$, one has $1 + B + g < B^{t+2} \leq q$ and $b\ell < q$. In the first bracket for R, the potentially negative term is absorbed by $\lambda B q^4 > 2z\lambda(1 + B + g)^4$. Both brackets are positive, so $R \geq 2$, $C > 0$, and $P \geq 2R^2 > R + 2$. The margin implies $K > 0$. If $h \leq 0$, then $1 \leq K \leq R + 1 < P$. The case $K = 1$ would imply $P - 1 \mid R$; for $2 \leq K < P$ the radicand lies strictly between $(PK - 1)^2$ and $(PK)^2$. Both contradict the hypotheses. Hence $h \geq 1$.

New Pell-orbit lemma. Under those hypotheses and (JR), $A \geq 4$, $D \geq A^2$, $F > 0$ and $G_1 > A > 1$. Since $F \equiv 1 \pmod{D}$ and $F \equiv 1 \pmod{J}$, the divisibility gives $H \equiv C \pmod{F}$. Thus

$$G_1 \equiv 1 \pmod{D}, \quad I \equiv 1 \pmod{D}, \qquad G_1 \equiv A \pmod{F}, \quad I \equiv D \pmod{F}.$$

Therefore D, F, I are positive and pairwise coprime. As DFI is a square, all three are squares. Put $d = G_1^2 - 1$, a positive nonsquare, and $x = \sqrt{I} > 0$. Even if $H < 0$, $x + H\sqrt{d} > 0$. The unit $\eta = G_1 + \sqrt{d} > 1$ has norm one, so some positive power is the identity modulo CF in the integer quadratic algebra. Multiplying $x + H\sqrt{d}$ by arbitrarily large such powers yields

$$x_n^2 - dH_n^2 = 1, \qquad H_n \equiv H \pmod{CF}, \qquad H_n \longrightarrow +\infty.$$

Choose $H_n > B_1$ and put $j_n = (H_n - B_1)/C > 0$. This preserves a_1 being square and $D_0 \mid T_0$, and leaves a_0, m unchanged. Thus the positivity guard on j can be deleted entirely.

3.3 Cofinal bases and square witnesses: two more deleted guards

The proof of `USys_of_mem` in pinned `Master3.lean`, lines 303–510, chooses an exponent $k = xy + \sum n_j + 1$ only to obtain $k \geq 1$, $xy < 2^k$, and $n_j < 2^k$ for a fixed quartic witness. The rest of the construction uses those inequalities, not that equality. Hence it works for every sufficiently large exponent k, including any prescribed parity. `NineReduce.reduced_of_mem` and `NineRatio.ratio_complete` transport that same base $b = 2^k$. At $x = 1$ we may therefore choose $k = \kappa + 2r$ large enough, and set $\varepsilon = 2^r$, giving $b = a\varepsilon^2$. Conversely any nonzero integer ε gives $b \geq a > y$, so the old ceiling coordinate $\varepsilon_0 = b - y - 1$ is nonnegative. This is a cofinal restriction of the base, not an unjustified square substitution for ε_0.

There is a second concrete freedom. In pinned `Ratio.lean`, `exists_B25core`, lines 434–442, the canonical witness is

$$\omega = 2^{2R+1-k}, \qquad U = N^2\omega.$$

The source first proves $k < 2R+1$ from $k < 2^k = b \leq R$, so natural subtraction does not truncate. Since our k is odd, $2R + 1 - k$ is positive and even: this ω is a positive square. `NineRatio.ratio_complete`,

lines 270–279, retains that same ω while adding i, j and changing ϕ. Thus $\omega = w^2$, $w \neq 0$, preserves completeness; soundness follows from $w^2 \geq 1$. There is no analogous asserted square property for s.

Consequently only $g, s \geq 1$, $\phi \geq 0$ and $m > 0$ require inequality guards. Nonvanishing of ε, w, i is sufficient for the other restrictions. These source-inspected strengthenings are ordinary proofs, not newly compiled formal theorems.

3.4 Why the finalizer has exactly the intended zeros

Every B'_j is 1 modulo 4 and coprime to C_{j-1}. If all $z_j \neq 0$ and C_* is a square, unique factorization forces $|B'_j|$ to be squares; their residue modulo 4 excludes negative squares. Thus $d_j \geq 0$, and $d_j \equiv 2 \pmod 3$ excludes zero. These are exactly the four required inequalities. Conversely each positive $d_j \equiv 2 \pmod 3$ is nonsquare. A Pell unit power gives $u_j^2 - d_j v_j^2 = 1$ with v_j divisible by $2|C_{j-1}|$. Taking $z_j = v_j/(2C_{j-1}) \neq 0$ makes every B'_j a square.

The polynomial $\Pi(t) = (t^3 - t - 1)(t^2 + 23)$ has no rational root but has a root modulo every nonzero integer. To see the latter, for an odd prime other than 23, either the cubic has a simple root, or its irreducibility makes Frobenius a three-cycle and its discriminant -23 a nonzero square. Thus one factor has a simple root and lifts. At 23 use the cubic root 3; at 2 use $-23 \equiv 1 \pmod 8$ and the standard square-root lifting. Apply the Chinese remainder theorem. For $D \neq 0$ the homogeneous expression

$$\mathcal{R}(D, T, v) = ((T + Dv)^3 - D^2(T + Dv) - D^3)((T + Dv)^2 + 23D^2)$$

is nonzero and satisfies

$$(D \mid T \ \&\ Z \neq 0) \quad \Longleftrightarrow \quad \exists v : \ ZD^5 \mid \mathcal{R}(D, T, v).$$

The forward direction applies intersectivity after $T = Dq$. For the converse, if $v_p(D) = \alpha > \beta = v_p(T)$, the two factors have valuations $3\beta, 2\beta$, contradicting divisibility by D^5. The case $Z = 0$ would contradict nonvanishing. This supplies exactly $D_0 \mid T_0$ and all nonzero conditions via $S \mid T$.

The new linear weight is valid even before recovering signs: $C_* \equiv 1 \pmod 4$ implies

$$|C_* + 1| > \sqrt{|C_*|}, \qquad \sqrt{|a_0|} \leq P^2 + K^2 + 1.$$

For negative $C_* = -t$, necessarily $t \geq 3$ and $(t-1)^2 - t = t^2 - 3t + 1 > 0$. Hence

$$|w_2| > \sqrt{|C_*|} + |w_1|\sqrt{|a_0|}.$$

For complex roots $r_0^2 = C_*, r_1^2 = a_0, r_2^2 = a_1$, multiplication gives

$$F_{14} = \prod_{\sigma \in \{-1,1\}^3} \big[X - S(\sigma_0 r_0 + \sigma_1 w_1 r_1 + \sigma_2 w_2 r_2)\big].$$

If $S = 0$, this is $T^8 \neq 0$. Otherwise a zero makes one encoded root sum rational. The strict absolute-weight bounds separate all sign choices on nonzero roots, including imaginary roots. Every Galois automorphism of their multiquadratic field must therefore fix each root, so all three roots are rational, hence integers. It follows that $S \mid T$ and C_*, a_0, a_1 are integer squares. Conversely these conditions yield a zero by taking $u = T/S + r_0 + w_1 r_1 + w_2 r_2$. Zero roots and repeated square classes cause no exception.

Combining this exact elimination with the square stack, homogeneous merger, recovered h, sign change of i, and Pell orbit for j recovers the original Jones witness. Conversely the cofinal-base and square-ω constructions, then the stack and finalizer, produce an integer zero from any quartic witness. The imported compiler and literal K215 calibration give

$$\mathrm{ZFC} \vdash \big(\nexists \mathbf{t} \in \mathbb{Z}^{14} : F_{14}(\mathbf{t}) = 0\big) \ \longleftrightarrow \ \mathrm{Con(ZFC)}.$$

3.5 Lower-degree alternative and exact complexity

For F_{16} keep $b = y + 1 + \varepsilon$, $\omega = w$ and use six stack discriminants

$$3\varepsilon + 2,\ 3g - 1,\ 3s - 1,\ 3w - 1,\ 3\phi + 2,\ 3m - 1, \qquad Z = i \prod_{r=1}^{6} z_r.$$

The same finalizer and signed linear weights apply. Its proof uses only the new j-orbit and weight improvement, not either cofinal specialization. The intermediate 15-variable construction restricts only b; its 1530 operations and degree $813056L + 4184$ are dominated by F_{14}.

The fast evaluator shares fifth powers, evaluates the background digit sum by the complete homogeneous polynomial h_4, and uses fixed multivariate Horner evaluation for sparse corrections. In particular, $1 + t + t^2 + t^3 + t^4 = 1 + (t^2 + t)(t^2 + 1)$ saves a gate when t^2 is shared. All scalar multiplications and squares are charged; fixed coefficient construction is excluded, as in the comparison convention. For F_{14},

R	$470L$	P	$1900L + 4$
K	$1900L + 5$	a_1	$49280L + 98$
m	$4442L + 14$	C_*	$4442L + 133$
S	$68750L + 152$	w_2	$8242L + 143$
A_2	$203264L + 688$	F_{14}	$813056L + 2752$

Here $\deg A_2 > \deg X^2, \deg A_0, \deg A_1$, so the successive leading terms of V, W, F_{14} are A_2, A_2^2, A_2^4. A nonzero symbolic diagonal leading coefficient, certified modulo 1000003 as a scalar times a power of the positive fixed a, proves these are exact degrees. For F_{16} one obtains $\deg A_2 = 101632L + 2502$.

The delivered generator and checker reproduce the DAGs, counts, and degrees. Diagnostics include 216 actual-coefficient modular circuit comparisons, 1000 eight-factor norm evaluations, 840 coprimality cases, 120 negative-to-positive Pell orbits, and intersectivity tests for every modulus 1 through 1200. These checks supplement, and do not replace, the universal proofs above.

4 Positive norms: degree 1061 and a 109-operation evaluator

All unknowns in this section are strictly positive. There are exactly twelve: $J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t$. Every other name is a polynomial abbreviation. All constructions use the complete source-30 Wang943 tuple (β, K_0, M_C, M_F), with $\beta = 2^{5^{10}} - 1$ and the shifted high mask M_F. The shared coefficient rebuild reproduces all four selected bytestrings.

Common interface. Write $\omega = w$, except in the lowest-degree family, where $\omega = w + q$. Define

$$q = \beta J + 1, \quad C_w = q - F - \alpha, \quad X = \omega q, \quad Y = sq^3, \quad E = XY,$$
$$k = \eta + \zeta, \quad c = kY + \eta, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H,$$
$$\ell = EkY, \quad A = 1 + \ell(\ell + k), \quad G = 1 + \Delta c^2, \quad b_0 = X + ac,$$
$$R = ((q - 1)(q - F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,$$
$$N_k = k - hE - R, \qquad N_t = (K_0 + \omega)C_w + q - F - z(q - 1),$$
$$I = ic^2, \quad D = 1 + \Delta I^2, \quad Q = (\Delta I)^2, \quad d = Q - 1,$$
$$v = cT, \quad b = c + RD, \quad e = Q(Dv^2 + b^2) - 1, \quad B_1 = 2Qvb.$$

Use one of

$$C_1 = E + 3(X + Y), \quad C_2 = E + 2(X + Y) + sq^2, \quad C_3 = E + 2(X + Y) + sq,$$

where C_3 is used only with $\omega = w + q$. Put

$$M = I(2T + RC_j ic), \qquad L = 1 + HM, \qquad W = Lb_0 + Ht.$$

The new norm is the following integer polynomial, with no additional witnesses:

$$\mathcal{N} = \prod_{g=\pm\sqrt{G},\, \tau=\pm\sqrt{A},\, f=\pm\sqrt{D}} \left[d(W - Lg - HM\tau)^2 - H^2 e + H^2 B_1 f\right].$$

The presentations are $N_k = N_t = 1$, $\mathcal{N} = 0$, and the single equation

$$P = N_k N_t (1 + \mathcal{N}) - 1 = 0.$$

A half-sized carrier and the new restriction. Source 24 used $L = 1 + 2HM$. Its uniform conjugate estimate $|y| < M\sqrt{A}$ remains valid here, where $dy^2 = e - B_1 f$. Hence all roots of the inner polynomial are $x = M\tau + y$ with $|x| < 2M\sqrt{A}$. Before any root recovery, the authentic weak interface gives $q \geq 16$, $X \geq 16$, $Y \geq 4096$, $E > R + 2$, $k > E$, and $c > kY$. Consequently $G > A$ and $b_0 = \ell + E\eta + Yc + X > \ell + k > \sqrt{A}$. For the proposed smaller carrier, the negative main-root branch would give

$$x = \frac{W + L\sqrt{G}}{H} = t + (M + H^{-1})(b_0 + \sqrt{G}) > 2M\sqrt{A},$$

a contradiction. If $\sqrt{G}$ were irrational, its negative conjugate would also be an inner root; thus the main root is a positive integer. Conjugation next forces τ rational, then y and f rational. The inherited rational Pell saturation lemma makes y integral. Reduction modulo H, using $L \equiv 1 \pmod{H}$, restores $g = b_0 + H\gamma$, with $\gamma > 0$. The unchanged signed-quotient compiler interface then supplies the accepting tableau. Conversely a canonical positive witness extends by $t = y + M\tau + L\gamma > 0$.

For the degree-1061 version, $Y/X = sq^2/(w + q) < sq$, so C_3 satisfies the same source-24 bound that previously used sq^2. Completeness is not assumed: after authentic root recovery, $X = \omega q = 2^R$ and $R > 3q$, so $\omega > q$. Thus $w = \omega - q$ is strictly positive. Soundness of $\omega = w + q$ is immediate. This coordinate change is not literal identity of polynomials at the same supplied w.

Cheaper trace–remainder evaluation. Source 30's evaluator applies to every carrier above. Let

$$\begin{aligned}
n &= (HM)^2 A, & d_n &= dn, & H_Q &= H^2 Q,\\
H_P &= H_Q D v^2, & H_U &= H_Q b^2, & H_e &= H_P + H_U - H^2, \quad j = 4H_P H_U,\\
T_r &= 2d(W^2 + L^2 G), & K_r &= d^2(W^2 - L^2 G)^2,\\
a_t &= T_r - 2(d_n + H_e), & b_t &= (d_n - H_e)^2 - j - K_r,\\
\bar{v} &= 16 d_n j - a_t(a_t T_r + 2b_t), & u_r &= b_t^2 - a_t^2 K_r.
\end{aligned}$$

Then

$$\mathcal{N} = K_r \bar{v}^2 - u_r(T_r \bar{v} - u_r).$$

Sharing H_Q saves one multiplication; the final Horner expression saves another. Comparing its two terms directly saves one further operation for a system. These are exact identities, including degenerate quadratic specializations. Also $\mathcal{N}$ is a square modulo 4, so the product equation forces $\mathcal{N} = 0$ and $N_k = N_t \in \{1, -1\}$; the inherited weak-sign compiler restores the positive unit sign.

Choice	Product degree	Product ops	System degree	System ops
Source 30, unchanged polynomial	1081	109	1072	104 / 103
C_1, half carrier	1069	117	1060	112 / 111
C_2, half carrier	1065	119	1056	114 / 113
C_3, shifted w	1061	120	1052	115 / 114

The second system count uses the proved shift $z_{\mathrm{old}} = z + 1$, replacing $N_t = 1$ by $(K_0 + \omega)C_w = F + z(q-1)$. All binary $+, -, \times$ gates, including multiplication by numerals, are charged; comparisons are free.

Exact degree and proof boundary. For the C_2 norm, the leading homogeneous form is

$$2^{46} i^{48} (\eta + \zeta)^{96} w^{74} s^{176} (\beta J)^{646} (\beta J - F)^{16},$$

of degree 1056. Four of the sixteen linear conjugate factors have degree 35, two have degree 36, and ten have degree 38; the multiplier is $d^8 H^{16}$. For C_3, the first four degrees drop to 34. Their complete leader contains

$$X_0^2 Y_0^2 (\eta + 2\zeta) + (\eta + \zeta) s^3 q_0^8 w,$$

which is nonzero; retaining the tied $-E_0^2 \eta$ term is essential. This proves norm degree 1052, and $N_k N_t$ adds nine degrees. The standalone proof supplies all separation estimates and leading forms. Independent exact conjugate products and complete finite-field ray checks corroborate the circuits, not the unbounded compiler theorem. These are ordinary reductions formalizable in ZFC, not a consistency proof, a global optimality claim, or a new kernel-checked endpoint.

5 A sharper equation in nine natural or positive unknowns

The new nine-variable polynomial has exact total degree

$$\boxed{92324 \cdot 5^{32} + 296 = 21495856344699859614062796.}$$

Its supplied integer arithmetic circuit uses **1505 binary operations** over $\mathbb{N} = \{0, 1, \ldots\}$, or **1500 operations** after the coordinate translation to strictly positive integers. The earlier source27 positive circuit still uses fewer operations (1480), but at much higher degree; the new form does not dominate it on cost alone. Source 30 used $120724 \cdot 5^{32} + 366$ and 1840 operations. The new construction retains source 30's complete compact943 coefficient tuple and 31-witness quartic, combines them with source 28's signed-divisor finalizer, and refines its majorants. This is an ordinary mathematical synthesis of the stated imported theorems, not a newly kernel-checked endpoint or a minimality claim.

The fixed instance. The four ports are the unchanged compact943 tuple (β, K_0, M_C, M_F), with native radix $2^{5^{10}}$ and $M_F = M_{F,\mathrm{native}} + \beta$. Let $r_1, \ldots, r_{22}$ be the literal quadratic residuals in `nine/source/30-quartic31/arithmetic/`. Their sum of squares after shifting all 31 positive coordinates by one is $P_{31}(x, \mathbf{n})$, independent of the dummy input x. It has degree four and $P_{31}(x, \mathbf{0}) \geq 1$ because residual 5 equals one at the all-ones tuple. Source 30's exact elimination of one register transfers the inherited compiler theorem to this parent.

Put $\nu = 31$, $L = 5^{32}$ and $c_\alpha = (4 - |\alpha|)! \prod_j \alpha_j! \, [\mathbf{x}^\alpha] P_{31}$. All 402 nonzero coefficient recipes are supplied. The recorded bit lengths of the four authentic ports give, for every recipe $c_\alpha = \sum_t a_t \boldsymbol{\kappa}^{\mathbf{e}_t}$,

$$|c_\alpha| + 1 \leq 1 + \sum_t |a_t| 2^{\sum_j e_{tj} \, \mathrm{bits}(\kappa_j)} < 2^{45933757}.$$

Thus $z = 2^{45933757}$ and $y = \sum_{|\alpha| \leq 4} (z + c_\alpha)(2z)^{L - \sum_j \alpha_j 5^j}$ give the explicit admissible Jones index. The baseline includes all 58,905 exponent positions. These are bounded coefficient recipes, not additional unknowns or an unspecified enumeration index.

The new majorants. Use the unchanged Jones packing and ratio expressions, fully listed in `nine/PROOF.md`, on the eight natural outer coordinates. In particular

$$A = R\sigma, \quad \sigma = Y(U+1), \quad r = bw + \phi, \quad C = 2R + r + 1, \quad H = (j+1)C - r,$$

$$D = (A^2-1)C^2 + 1, \quad E = i(4A-5)DC^2, \quad F = (A^2-1)E^2 + 1, \quad G = FDC + F - A(F-1).$$

Source 28's unconditional congruence $R \equiv (b-1)\ell \pmod{q}$, with $0 < (b-1)\ell < q$, proves $R \neq 0$. Consequently $|A| \geq 2$, $P \geq 2$, $K \geq 1$, $AE \geq 0$ and $D, F \geq 1$; no margin or square condition is assumed. Replace the earlier majorants by

$$\begin{aligned} M_D &= AC + \sigma(r+1)^2 + 1, & M_F &= AE + 1, \\ T_{CH} &= CH + r^2, & M_I &= F(DT_{CH} + 2H^2 + A^2) - A^2 + 1, \\ V_0 &= PK + 1, & V_1 &= M_D M_F M_I + 1. \end{aligned}$$

Indeed $M_D - AC > 0$ and

$$M_D + AC = \sigma\{4R^2 + 2R(r+1) + (r+1)^2\} + 1 > 0,$$

so $M_D \geq |AC| + 1 \geq \sqrt{D}$. Also $M_F \geq \sqrt{F}$, and

$$T_{CH} + CH = C^2 + (C-r)^2 + 2jC^2 \geq 0, \qquad T_{CH} - CH = r^2 \geq 0.$$

Thus $T_{CH} \geq |CH|$. The decomposition of G, together with $|H| \leq H^2$ and $|AH| \leq A^2 + H^2$, gives $M_I \geq |GH| + |H| + 1 \geq \sqrt{|(G^2-1)H^2 + 1|}$. For $a_0 = (P^2-1)K^2 + 1$ and $a_1 = DF((G^2-1)H^2 + 1)$, we obtain $V_t \geq 2$ and $\sqrt{|a_t|} \leq V_t - 1$. The proof includes negative R, $C = 0$ and $H = 0$.

The signed finalizer. Let $u = JF \neq 0$, $v = QF + J(H - C)$, with $J = 4A - 5$, and let m be the unchanged Jones margin. Put $M = Q^2 + (H-C)^2 + 1$; then $|v/u| \leq |Q| + |H - C| \leq M - 1$. With the ninth unknown $n \in \mathbb{N}$,

$$\Omega = u(2m-1), \qquad \Theta = u(n+M) + v - \Omega(2M + V_0V_1),$$

$$\mathcal{F}_9 = \big(\Theta^2 - \Omega^2(a_0 + V_0^2 a_1)\big)^2 - 4\Omega^4 V_0^2 a_0 a_1.$$

Source 28's four-factor radical argument proves that $\exists n \in \mathbb{N} : \mathcal{F}_9 = 0$ is equivalent to a_0, a_1 being squares, $u \mid v$ and $m > 0$, for either sign of u. For clarity, a vanishing factor yields integer roots $r_t^2 = a_t$: if $s = r_0 + V_0 r_1 \in \mathbb{Q}$ is nonzero, use $2sr_0 = s^2 + a_0 - V_0^2 a_1$; if $s = 0$, the bounds force $a_1 = a_0 = 0$. Then $Z = 2M + V_0V_1 + r_0 + V_0 r_1 \geq 2M + 1$, and $n + M + v/u = (2m-1)Z$ gives soundness. Conversely choose $n = (2m-1)Z - M - v/u \geq 2$. The unchanged Jones reduction, parent compiler and literal K215 calibration therefore give, provably in ZFC,

$$\big(\nexists \mathbf{x} \in \mathbb{N}^9 : \mathcal{F}_9(\mathbf{x}) = 0\big) \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

Why the degree drops. The leading terms of V_1^2 and a_1 now cancel. This is resolved by an exact telescoping certificate, not a naive gate-degree estimate. With $t = \sigma(r+1)^2$,

$$M_D^2 - D = 2ACt + t^2 + 2AC + 2t + C^2$$

has degree $1110L + 8$ and diagonal leader 4. In

$$\begin{aligned} V_1^2 - a_1 = (M_D^2 - D)M_F^2 M_I^2 + D(M_F^2 - F)M_I^2 \\ + DF(M_I^2 - I) + 2M_D M_F M_I + 1, \end{aligned}$$

the first term uniquely dominates: its degree is $24170L+72$ and leader 2^{46}. Writing $E_0 = u(n+M)+v-2\Omega M$, the inner norm is exactly

$$E_0^2 - 2E_0\Omega V_0 V_1 + \Omega^2 V_0^2(V_1^2 - a_1) - \Omega^2 a_0.$$

The third term uniquely dominates with degree $46162L + 148$ and leader 2^{98}. The outer square therefore has degree $92324L + 296$ and leader $2^{196} > 0$; the final subtracted term has smaller degree.

Executable representation and checks. The shared radix-five ladder and complete-homogeneous h_4 identity from source 27 evaluate the 58,905-position digit baseline without expanding it. Exact coefficient aliases leave 78 nonliteral coefficient leaves besides z, y. The identity $1+t+t^2+t^3+t^4 = 1+(t^2+t)(t^2+1)$, deterministic multivariate Horner ordering and further distributive rewrites give the stated operation count. The supplied standard-library generator and verifier rebuild all 402 coefficient recipes, prove the complete sparse digit identity, check the actual coefficient hashes and index bound, audit the cancellation-free degree circuit and compare actual-instance evaluations modulo four primes. Finite tests supplement the universal proof; they do not decide solvability or recheck the complete semantic chain in a proof-assistant kernel.

6 Integer-domain synthesis

Two combinations improve different parts of the integer frontier. All their coefficient ports select the same complete compact Wang943 tuple, with $B = 2^{5^{10}}$ and the shifted high-mask port. Historical source bindings are not spliced into these new instances.

A quartic in fifty unrestricted integers. Source 30 removes the positive packing register G by the identity

$$L\big(q(q-F)-C\big) - \big((q-F)(N-q)-CL\big) = (q-F)\big(q(L+1)-N\big).$$

This identity is equally valid in source 27's signed odd-scale lift. Deleting G there, while retaining $i = 2r+1$, $s = 2v+1$, $w = u^2$ and six three-variable positive guards, gives fifty unrestricted integers and 29 quadratic equations. Restoring $G = q(q-F)-C$ is unique and precedes every sign-recovery argument; thus no positive-s assumption is imported prematurely. The sum of squares has exact degree four, since its τ^4 coefficient is one. Its comparison system has 111 binary operations, and the explicit quartic evaluator has 197. The expanded polynomial has 414 symbolic terms, including coefficient ports.

A cheaper degree-1936 polynomial in twenty-four integers. Keep the signed ratio-norm notation $I = ic^2$, $v = cT$, $b = c + RD$, $G = 1 + \Delta c^2$. The improved weight is

$$M = 2(5Rac\,iI + v^2), \qquad L = 1 + HM.$$

The inherited sign-safe bound strengthens to

$$\frac{|y'|}{\sqrt{G}} < iI + v^2 + 2 + 4Rac\,iI < 5Rac\,iI + v^2 = M/2,$$

because $(Rac-1)iI > 2$ already holds before root recovery. Both signs of s, i, H, and the $w = 0$, $T = 0$ boundaries, remain covered. With $S = X + ac + Ht$, take the eight-factor norm of

$$d(S - Lu - 2HM\tau)^2 - H^2 e + H^2 B_{\mathrm{aux}} f, \qquad u^2 = G, \quad \tau^2 = A, \quad f^2 = D.$$

Source 30's trace/product remainder method evaluates this norm. Sharing $HQ = H^2Q$ and using a Horner final norm further reduces its circuit. The product $N_kN_t(1+N)-1$ now costs 161 operations at exact degree1936. The three-equation presentation costs 156 at maximum degree1920, using a direct final comparison; its SOS costs 164 at degree3840. The uniform leading certificate is

$$N_* = (99 \cdot 899)^4 d_*^8 H_*^{16} b_*^{16}, \qquad \deg N = 1920.$$

This replaces the previous degree1936 evaluations at 167/168 operations.

Verification and scope. The generators deliver every binary gate and every quadratic residual. Independent checks establish the packing syzygy, the quartic expansion, 208 rank-eight conjugate-product comparisons, and exact finite-field degree certificates using the actual coefficient tuple. Full sign proofs and source provenance are in `lowdegree/PROOF.md`. The compiler and calibrated K215 endpoint remain inherited mathematical dependencies. These constructions assert neither consistency nor global minimality, and are not new end-to-end proof-assistant certificates.

7 Validation, attribution, and limitations

Every new family includes a generator, explicit arithmetic circuits or sparse residuals, a mathematical proof, and replay checks. The final verification receipt records the actual completed stages and byte-identical regenerated artifacts. Separate independent reviews address the domain-restoration lemmas and root separation. Corruption controls are designed to reject a changed coefficient payload, a changed circuit, and a missing source dependency. They do not test or decide whether a genuine ZFC-refutation witness exists.

The full source comparisons and proof notes identify inherited facts separately from new arguments. The symbolic identities establish polynomial equality where that is claimed; modular evaluations are diagnostics, not proofs of equality. Exact-degree claims include nonzero leading-term certificates rather than merely structural upper bounds. All original source snapshots retain their provenance and original tuple metadata; only newly generated packets select the shared binding.

This archive does not replace an endpoint build. The published Lean/Rocq K215 endpoint and the inherited native-window/Pell compiler are used through their identified ordinary mathematical calibration. New cofinal-base and finalizer arguments have not been compiled into either proof assistant. Package 27's earlier licence omission is not repeated: the Wade licence and K215 notices are included alongside the repository licence and source attribution. Source scripts or snapshots omitted during repository intake are not misidentified as defects of the original sealed archives.

P67 remains the lowest-operation single equation in the reviewed collection. The older low-degree and low-operation systems remain useful when they are not dominated. Fourteen unrestricted integers is a concrete improvement over the reviewed seventeen-integer construction, not a global lower bound or a claim to supersede abstract representation theorems. All claims concern the stated domains and the coherent fixed coefficient tuple.